

Saransh Agrawal

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Education

Texas A&M University

Aug 2023 – May 2025

Master of Science in Data Science: GPA: 3.7/4.0

College Station, TX

- Research Assistant at *FLAIR Lab* - Responsible NLP, Safety, Robustness.

- Teaching Assistant, Dept. of Statistics (Mentored 50+ students)

Manipal Institute of Technology

July 2017 – Aug 2021

Bachelor of Technology in Control Systems Engineering

Manipal, IN

Technical Skills

Languages: Python, Go, C/C++, SQL, Shell/Bash.

ML/AI Frameworks: PyTorch, Transformers, TensorFlow, Scikit-Learn, OpenAI, HuggingFace, vLLM.

MLOps & Infrastructure: Docker, Kubernetes, Apache Kafka, AWS (EC2, S3, SageMaker), SLURM, Jenkins CI/CD, Git.

Data & Storage: PostgreSQL, MongoDB, Elasticsearch, Apache Spark, Pandas, NumPy.

Specializations: Large Language Models, Multimodal AI, Model Optimization, Distributed Training, ML Pipeline Engineering.

Publications

Saransh Agrawal and Kuan-Hao Huang. SHA256 at SemEval-2025 Task 4: Selective Amnesia – Constrained Unlearning for Large Language Models via Knowledge Isolation. In *SemEval 2025 at ACL*.

Projects

- **VLM-LongContext: Efficient Long-Context Reasoning in Multimodal LLMs** — *Python, PyTorch, LLaVA, HuggingFace*
 - Built ML optimization framework reducing memory consumption by **75%** for long-context multimodal reasoning using Streaming LLM and Cache Merging.
 - Implemented **image chunking with descriptive generation** enabling processing of 100+ images vs 6-8 baseline limit on MileBench dataset.
 - Benchmarked LLaVA-1.5, MiniGPT-4, and LLaVA-OneVision models, achieving optimal memory-performance tradeoffs.
- **Preference Vector Framework for LLM Alignment** — *Python, PyTorch, Transformers, DPO*
 - Collaborated with National Taiwan University (5-person international team) to design modular DPO framework.
 - Enabled controllable post-training alignment by dynamically merging preference vectors at test time with user-adjustable scaling coefficients.
 - Outperformed Safe-RLHF baselines with 4x faster training. *Submitted to COLM 2025 (preprint)*.
- **Satellite-based Crop Monitoring System (SCMS)** — *Python, FastAPI, Docker, React.js, MongoDB, PostgreSQL*
 - Led 4-person team building data ingestion, transformation, and serving pipelines for satellite imagery processing.
 - Integrated PlanetScope satellite API for automated imagery acquisition with cloud filtering and AOI-specific preprocessing.
 - Deployed LSTM models achieving $R^2 = 0.81$ for cotton yield prediction. *Presented at ASA Conference 2024*.
- **SLURM GPU Job Tutorial & MMLU Benchmarking** — *Python, PyTorch DDP, SLURM, Bash Scripting*
 - Created comprehensive tutorial for distributed LLM benchmarking across single/multi-GPU and single/multi-node configurations.
 - Built fault-tolerant distributed execution framework using PyTorch DDP with automated SLURM job templates.

Work Experience

Software Developer, ML Team

July 2021 – July 2023

Viewzen Labs

Bangalore, India

- Spearheaded development of distributed AutoML platform (MLaaS), adopted by 12+ enterprise clients, processing 800K+ requests monthly.
- Developed system architecture orchestrating microservices through REST APIs and Kafka Streams, collaborating with 8-person cross-functional team.
- Built Python library implementing AutoML logic using Hyperopt, Scikit-Learn, reducing ML development time by 80%.
- Created RESTful Python microservices with Docker containerization. Mentored 2 junior developers on API design.
- Enhanced processing speeds by 20% through Cython optimization and multiprocessing for high-volume data transformation.
- Led predictive modeling initiative for 45,000+ women (300+ features), culminating in medical desktop application.

Machine Learning Engineer Intern

Feb 2021 – June 2021

Centre for Digital Financial Inclusion

Delhi, India

- Transformed raw audio signal to its MFCC components using custom TensorFlow layers.
- Implemented RNN, GRU and LSTM architectures for text prediction at character, word and phoneme level.
- Utilized DeepSpeech transfer learning on Indian Accent dataset achieving 0.10 CER and 0.17 WER with KenLM language model. Added T5 Transformer to generate SQL queries.